Alternative Assessment Tool (AAT) Report on

"Design and Verification of Combinational and Sequential Digital Modules using Verilog HDL"

Submitted in partial fulfilment of the requirements of the Alternative Assessment Tool (AAT) for the course

ADVANCED DIGITAL AND LOGIC DESIGN (ADLD)

Course Code: 23ECSPE1AD

Submitted by

Mallika Gourish Hegde 1BM24EC100

Submitted to

Prof. Ajaykumar Devarapalli

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

BMS College of Engineering

Bull Temple Road, Basavanagudi, Bengaluru-560019

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INTRODUCTION TO CADENCE

Cadence Design Systems :

Cadence Design Systems is a leading Electronic Design Automation (EDA) company that provides advanced software tools for designing, analyzing, simulating, and verifying integrated circuits (ICs) and electronic systems. Cadence tools are widely used in the semiconductor and VLSI industries for digital and analog circuit design, RTL development, functional verification, logic synthesis, physical design, and timing analysis.

In digital design, Cadence provides tools such as Cadence Virtuoso, Genus, Innovus, and Xcelium, which support different stages of the IC design flow. Virtuoso is primarily used for analog and custom IC design, Genus for logic synthesis, Innovus for physical implementation, and Xcelium for functional simulation and verification.

Cadence tools help designers identify design errors, analyze circuit performance, optimize power, performance, and area (PPA), and verify the functionality of a design before manufacturing. These tools play an important role in modern VLSI design by reducing development time, improving design accuracy, and ensuring reliable integrated circuit implementation.

APPLICATIONS OF CADENCE IN CIRCUIT DESIGN

Cadence plays an important role in designing and analyzing different types of electronic circuits. It is commonly used for designing analog circuits, digital circuits, combinational logic circuits, sequential circuits, and mixed-signal systems. It also supports the development of complex integrated circuits used in communication systems, processors, control systems, and other electronic applications.

4-Bit Barrel Shifter

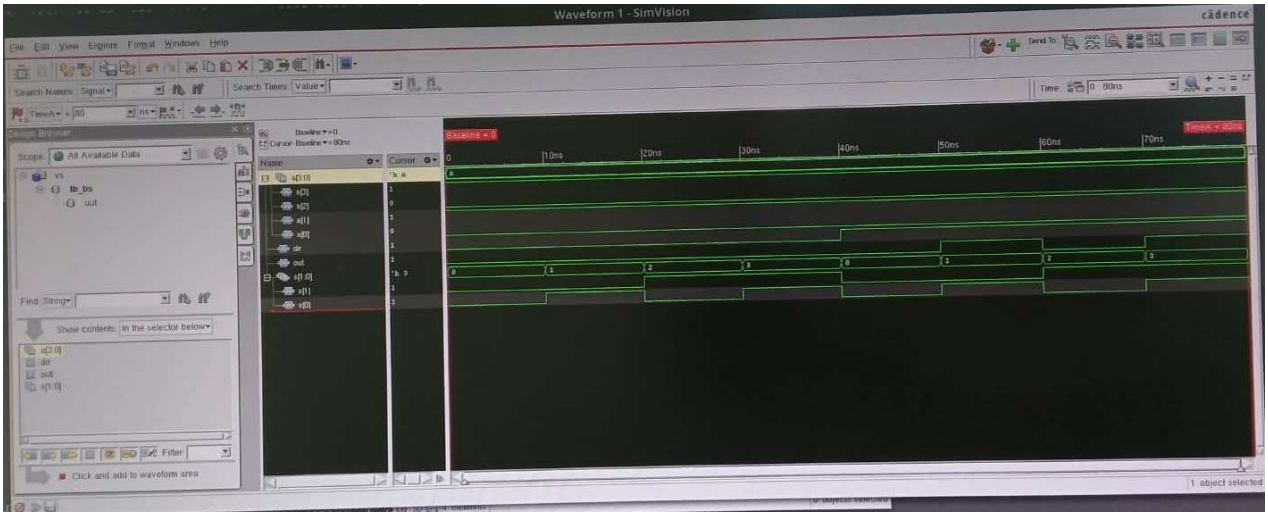
- **Verilog code :**

```
module barrel_shifter_4bit (
input [3:0] data_in, // Input data
input [1:0] shift_amt, // Shift amount (0 to 3)
input direction, // Shift direction: 0=Left, 1=Right
output [3:0] data_out // Shifted output );
always @(*) begin
case (direction)
  1'b0: // Left shift
     case (shift_amt)
        2'b00: shifted = data_in;
        2'b01: shifted = {data_in[2:0], 1'b0};
        2'b10: shifted = {data_in[1:0], 2'b00};
        2'b11: shifted = {data_in[0], 3'b000};
        default: shifted = data_in; endcase 1'b1: // Right shift
    case (shift_amt)
       2'b00: shifted = data_in;
       2'b01: shifted = {1'b0, data_in[3:1]};
       2'b10: shifted = {2'b00, data_in[3:2]};
       2'b11: shifted = {3'b000, data_in[3]};
       default: shifted = data_in;
  endcase
default: shifted = data_in;
endcase
end
    assign data_out = shifted;
endmodule
```

- **Test bench :**

```
module tb_barrel_shifter_4bit_ifelse;
 reg [3:0] data_in;
reg [1:0] shift_amt;
reg direction;
wire [3:0] data_out;
barrel_shifter_4bit_ifelse uut ( .data_in(data_in), .shift_amt(shift_amt), .direction(direction),
.data_out(data_out) );
initial begin
  direction = 0; data_in = 4'b1011; shift_amt = 0; #
  shift_amt = 1; #5
  shift_amt = 2; #5
  shift_amt = 3; #5
  direction = 1; shift_amt = 0; #1
  shift_amt = 1; #5
  shift_amt = 2; #5
  shift_amt = 3; #5
  $finish;
end
endmodule
```

- **Output waveform :**



Jhonson counter

- **Verilog code :**

```
module johnson_counter(
    input wire clk,
    input wire reset,
    output reg [3:0] q
);

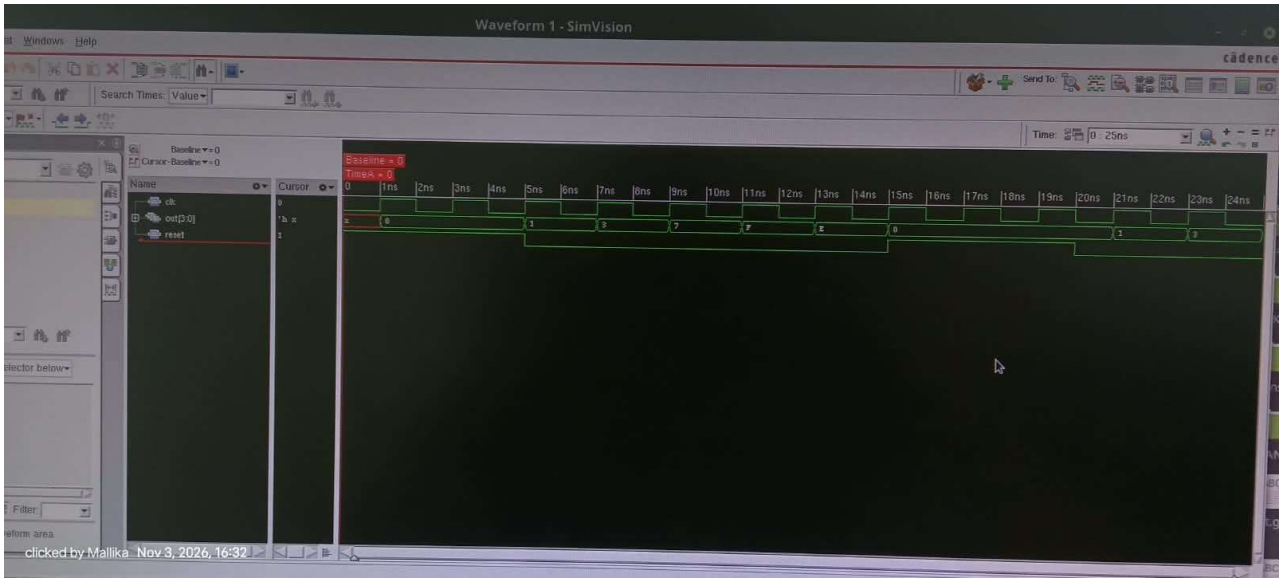
    always @(posedge clk) begin
        if (reset) begin
            q <= 4'b0000;
        end else begin
            q <= {~q[0], q[3:1]};
        end
    end

endmodule
```

- **Test bench :**

```
module johnson_counter_tb;
reg clk;
reg reset;
wire [3:0] q; // Instantiate the Johnson counter
johnson_counter uut ( .clk(clk), .reset(reset), .q(q) ); // Clock generation: 10ns period
initial begin
 clk = 0; forever #5 clk = ~clk; end // Test sequence
initial begin
$dumpfile("j_counter.vcd")
$dumpvars // Initialize
reset = 1; #10;
reset = 0; #10; // Apply reset again mid-run
reset = 1; #10;
reset = 0; #50;
$finish;
end
endmodule
```

Output waveform :



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Mallika Gourish Hegde

USN: 1BM24EC100

Github repository link :

https://github.com/MallikaGHegde/Digital-Logic-Design-ADLD-AAT-.git